0.1 General Linear Group

Definition 0.1.1. Define a *General Linear Group* over a field F as:

$$\mathrm{GL}_n(F) \overset{\text{def}}{=} \{A \in \mathcal{A}_{n\times n}(F) \mid \det A \neq 0\}$$

And, a *Special Linear Group* as:

$$\mathrm{SL}_n(F) \overset{\text{def}}{=} \{A \in \mathcal{A}_{n\times n}(F) \mid \det A = 1\}$$

The property of determinant allows above set are Group under the matrix multiplication.

Lemma 0.1.2. A map

$$\varphi : \mathrm{GL}_n(F) \to F^\times : A \mapsto \det A$$

is an Onto Homomorphism with $\ker\varphi = \mathrm{SL}_n(F)$. Therefore,

$$\mathrm{GL}_n(F)/\ker\varphi = \mathrm{GL}_n(F)/\mathrm{SL}_n(F) \cong F^\times$$

Proof. Well-Defined is obtained by the determinant was obtained by operation in the given field.
And, surjectiveness is given by $\det \mathrm{diag}(1, \ldots, 1, a) = a$ for all $a \in F$.
And homomorphism is obtained by the determinant preserves matrix multiplication, the kernel is trivial.
Finally, the First-Isomorphism Theorem gives the result. □

0.1.1 General Linear Group over the Finite Field

Proposition 0.1.3. Let $\mathbb{F}_q$ be a finite field of order q. Then,

$$\begin{aligned} \mathrm{GL}_n(\mathbb{F}_q) &\text{ has order } (q^n-1)(q^n-q^1)\cdots(q^n-q^{n-1}) \\ \mathrm{SL}_n(\mathbb{F}_q) &\text{ has order } \frac{1}{q-1}|\mathrm{GL}_n(\mathbb{F}_q)| \end{aligned}$$

Proof. Note: Clearly, $|M_{n\times n}(\mathbb{F}_q)| = q^{n^2}$.
To form an invertible matrix, we choose n linearly independent columns.
Choice of the first column as non-zero: there are $q^n - 1$ possibilities.
And, the second column can be any vector except a multiple of the first, then $q^n - q$ possibilities.
Successively, choose the third as any vector but not a linear combination of the first and second, then there are $q^n - q^2$
(More specifically, q^2 is the number of cases of $a_1u_1 + a_2u_2$).
Inductively, we obtain that

$$|\mathrm{GL}_n(\mathbb{F}_q)| = \prod_{k=0}^{n-1}(q^n - q^k)$$

And order of $\mathrm{SL}_n(\mathbb{F}_q)$ is obtained by the above lemma. □

Note: Since the properties of the finite field, the order q must be a power of a prime.